

Risk-Based Borrowing Limits in Credit Card Markets

ONLINE APPENDICES

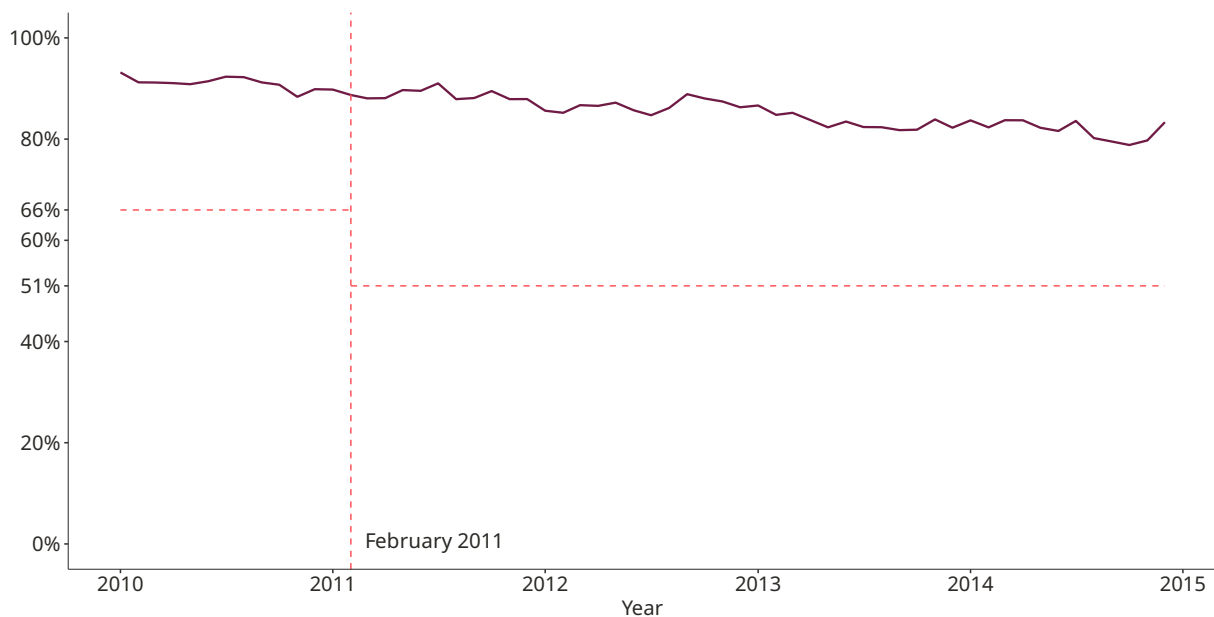
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September 13, 2026

A Additional Figures and Tables

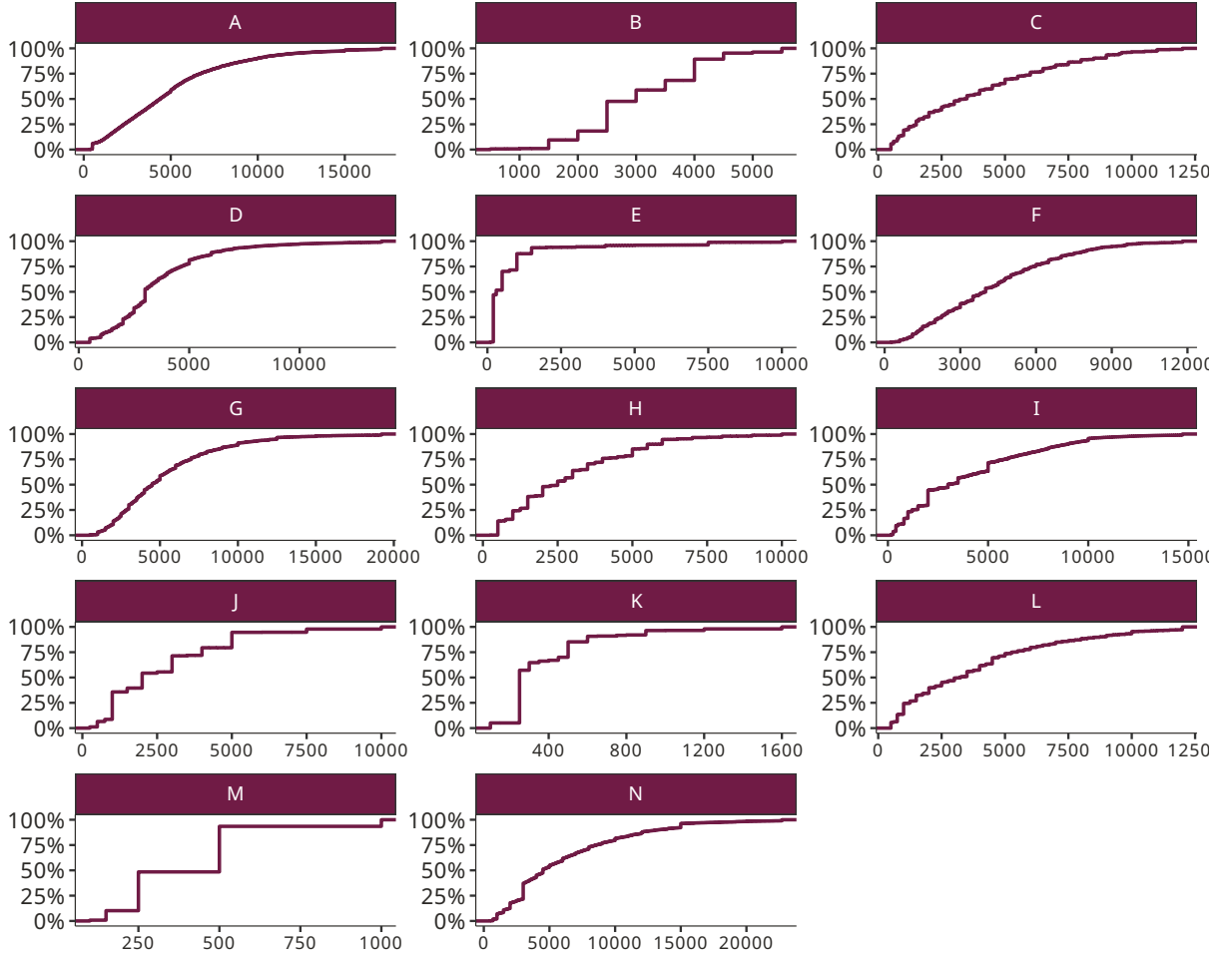
A.1 Figures

FIGURE A.1. PERCENTAGE OF ORIGINATIONS THAT OBTAIN ADVERTISED APR



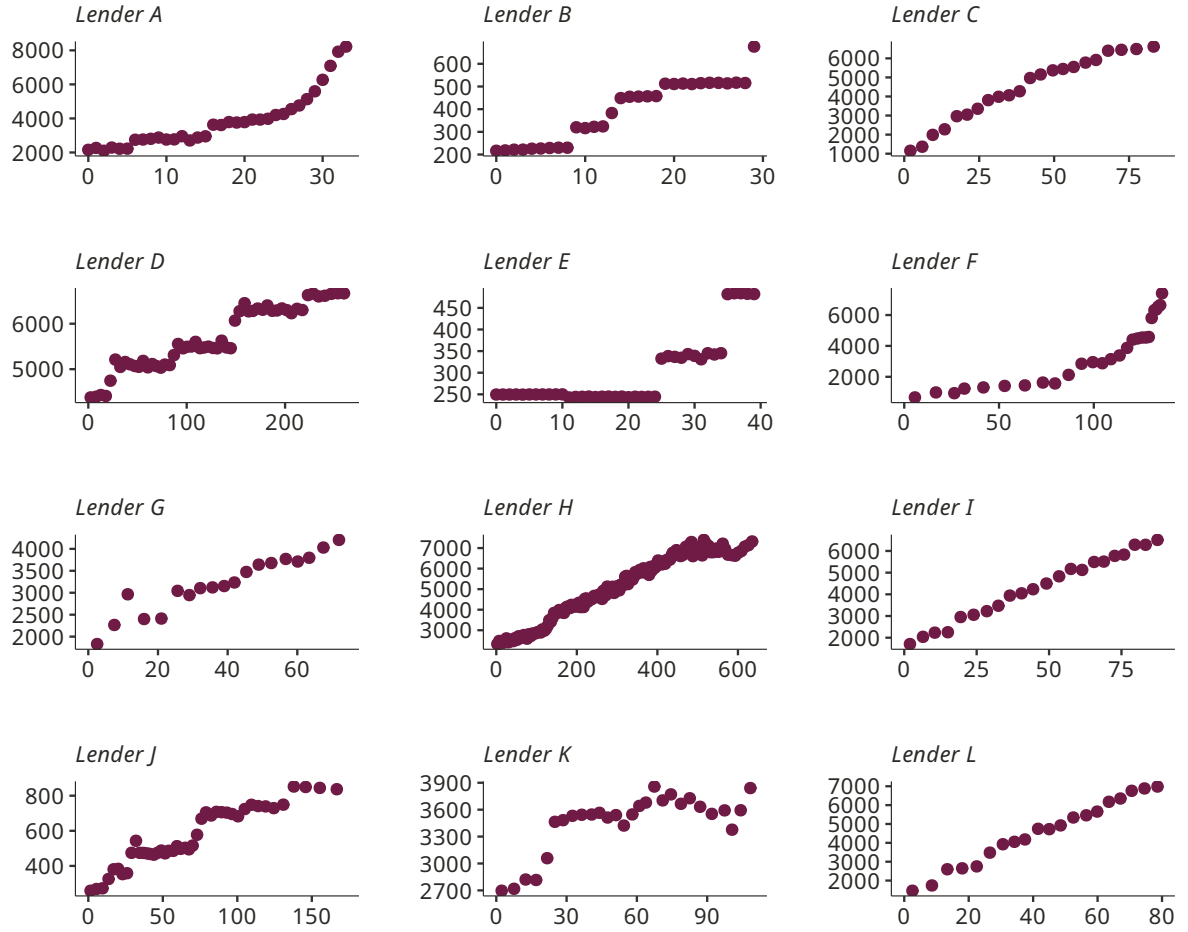
Notes: The horizontal dashed lines indicate the legal requirement on the percentage of customers obtaining at most the advertised APR. The vertical dashed line indicates February 2011, the month in which the threshold was changed from 66% to 51%.

FIGURE A.2. EMPIRICAL CDFs OF CREDIT LIMITS AT ALL LENDERS



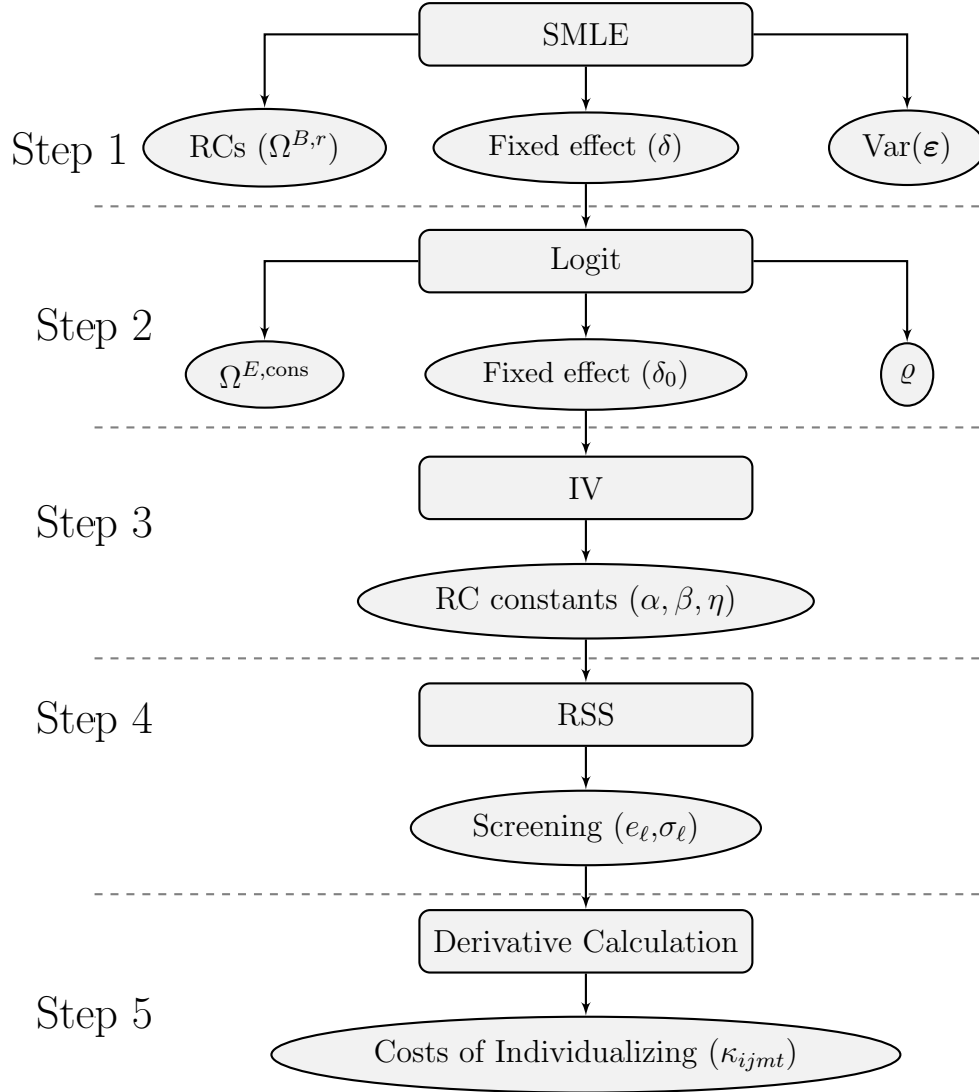
Notes: I scramble lenders' identities to preserve anonymity, so labels do not necessarily match the identities in other tables and figures. I include store cards as well as the 13 other lending entities here.

FIGURE A.3. MEAN CREDIT LIMITS ACROSS LENDERS' CREDIT SCORES



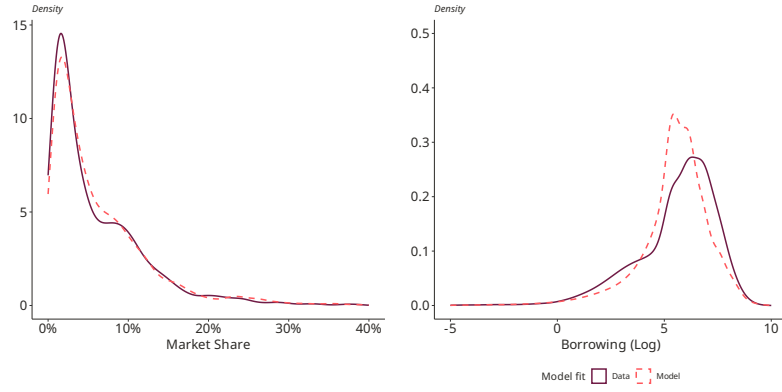
Notes: Plots are for the year 2013. I scramble lenders' identities to preserve anonymity, so labels do not necessarily match the identities in other tables and figures. Credit score scales differ across lenders, so they cannot be compared. Credit scores are not available for one lender in this year.

FIGURE A.4. STEPS OF MODEL ESTIMATION



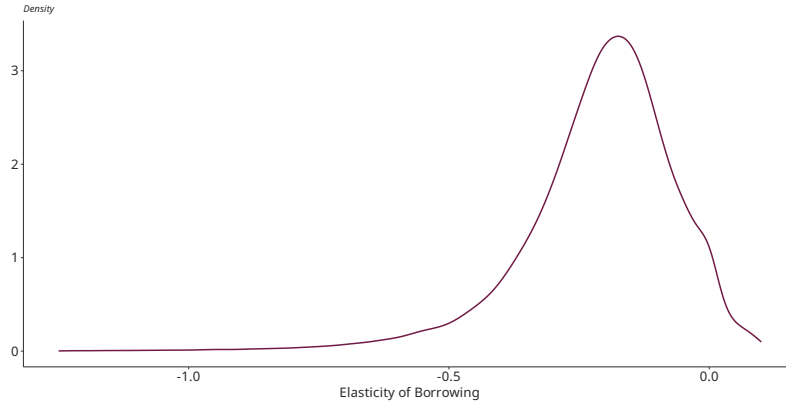
Notes: Step 1 refers to simulated maximum likelihood estimation of the demand parameters, using data on those who revolve. Step 2 refers to maximum likelihood estimation of the parameters governing the transaction utility. Step 3 refers to instrumental variables estimation of the parameters inside the fixed effects δ_{jmt} , i.e., α^r, β^r and η^r for $r \in \{E, B\}$. Step 4 refers to screening technology estimation. Finally, step 5 refers to estimating the costs to lenders of individualizing interest rates.

FIGURE A.5. DENSITIES OF MARKET SHARES AND BORROWING IN DATA AND MODEL



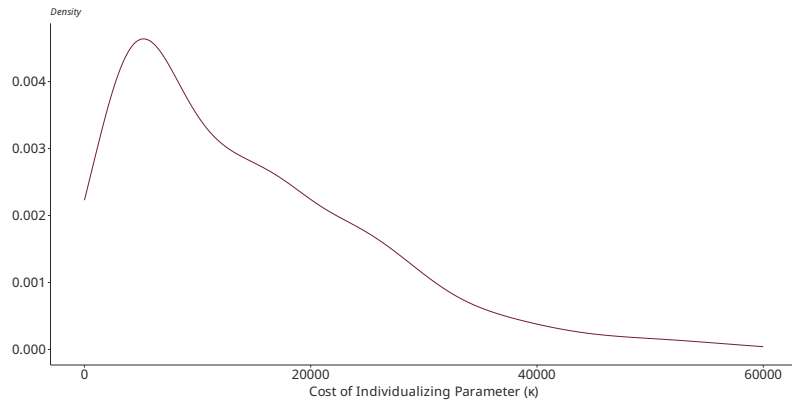
Notes: The left panel plots the distribution of market shares; the right panel plots the distribution of log borrowing. Each panel overlays the data and the model-implied distribution.

FIGURE A.6. DISTRIBUTION OF THE INTEREST RATE ELASTICITY OF BORROWING



Notes: Equation (C.3) defines the borrowing-choice elasticity of demand.

FIGURE A.7. DISTRIBUTION OF COSTS OF INDIVIDUALIZING INTEREST RATES, κ_{ijmt}



A.2 Tables

TABLE A.1. SUMMARY STATISTICS

Variable	Mean	SD	25%	50%	75%
<i>Panel A: Cardholder Features At Origination</i>					
Net Monthly Income (£)	2011.50	4875.90	1049.33	1517.33	2229.56
Age	42.87	14.83	31.00	41.00	53.00
Female	0.52	0.50			
Employed	0.76	0.43			
Homeowner	0.57	0.50			
Existing Customer	0.46	0.50			
Multiple Users	0.08	0.26			
Distribution: Branch	0.32	0.46			
Distribution: Online	0.53	0.50			
Distribution: Post or Telephone	0.15	0.36			
<i>Panel B: Card Features At Origination</i>					
Credit Limit (£)	3378.36	3138.53	1000.00	2500.00	5000.00
Purchase APR (%)	21.56	7.66	16.90	18.90	23.95
Get Advertised APR	0.83	0.37			
Purchase Promo Length	3.55	4.70	0.00	3.00	6.00
Balance Transfer Promo Length	9.23	8.73	0.00	9.00	15.00
Balance Transfer	0.28	0.45			
<i>Panel C: Statement</i>					
Credit Limit (£)	3849.30	3325.63	1250.00	3000.00	5200.00
Purchase APR (%)	18.30	9.54	15.80	17.90	21.90
Closing Balance (£)	1139.33	1864.64	0.00	357.12	1456.70
Repayment (£)	203.66	595.52	0.00	26.59	129.05
Total Interest (£)	8.54	20.15	0.00	0.00	7.57
Direct Debit	0.34	0.47			
Up-to-Date	0.95	0.22			
<i>Panel D: Card-Month Characteristics</i>					
Yearly Minimum Income Threshold (£)	5967.27	10798.66	0.00	3000.00	7500.00
Annual Fee (£)	9.25	34.95	0.00	0.00	0.00
Advertised APR (%)	23.39	11.35	16.90	18.90	29.80
Grace Period	28.78	11.96	25.00	25.00	26.00
Yearly Funding Rate (%)	2.28	0.98	1.59	2.29	2.81
Per-Origination Operational Cost (£)	2.66	1.93	1.69	2.39	3.38
Per-Origination Overhead Cost (£)	3.03	1.36	2.22	2.66	4.18
Airmiles	0.06	0.23			
Affinity	0.21	0.41			

Notes: Any variable with (£) is in 2015 Pounds Sterling (GBP). Any variable without percentiles is a categorical (dummy) variable. The means of categorical variables across categories may not add to 1 because of rounding.

Panel A Notes: Unit of observation is the credit card origination (it_0). “Net Monthly Income” is net of all taxes. “Homeowner” is equal to one if the individual owns a house (with a mortgage or without) at origination. “Existing customer” is equal to one if the individual had any other financial product with the lender at the point of origination. “Employed” does not include self-employment. “Multiple users” is equal to one if the individual created multiple instances of the card at origination.

Panel B Notes: Unit of observation is the credit card origination (it_0). Promotional deal lengths are measured in months. “Balance Transfer” is equal to one if the originator transferred a balance from another card onto this newly originated card at origination. “Get Advertised APR” is a dummy equal to one if the individual obtains the APR advertised in the promotional materials.

Panel C Notes: Unit of observation is the statement-month (it). “Closing Balance” includes purchase, cash advance, money transfer, and balance transfer balances. “Total Interest” includes purchase, cash advance, money transfer, and balance transfer interest. “Direct Debit” is equal to one if some form of direct debit was used to pay this statement-month. “Up-to-Date” equals one if the individual repays at least the minimum payment this statement-month.

Panel D Notes: Unit of observation is the card-month (jt). “Yearly Minimum Income” is gross of tax. “Grace Period” is the number of days between the end of a billing cycle and the payment deadline. “Operational” and “Overhead” costs are per-origination. Reward variables all equal 1 if the card-month offers the reward and 0 otherwise.

TABLE A.2. CREDIT SCORE, INTEREST RATE, AND CREDIT LIMIT VARIATION BY LENDER

Bank	Credit Score	Interest Rate				Credit Limit				(10) Share (%)
	(1) Within	(2) C. of V. (%)	(3) 75/25	(4) 90/10	(5) Within	(6) C. of V. (%)	(7) 75/25	(8) 90/10	(9) Within	
A	97.66	11.02	1.19	1.32	20.45	77.89	3.28	8.98	88.53	2.21
B	89.41	15.14	1.25	1.39	45.61	79.49	4.57	11.74	77.89	8.33
C	51.88	22.41	1.29	1.59	18.62	84.23	4.45	16.18	71.09	21.96
D	95.41	14.12	1.02	1.66	23.13	74.37	3.87	9.76	73.92	3.18
E	95.79	10.50	1.09	1.27	49.50	75.57	3.16	10.82	81.23	7.77
F	-	11.53	1.11	1.21	-	59.18	2.65	6.08	-	6.02
G	46.22	12.27	1.06	1.32	2.06	164.12	4.71	9.98	25.13	8.54
H	93.87	6.56	1.11	1.15	0.00	65.65	2.07	5.18	99.48	11.44
I	96.77	22.62	1.53	1.77	67.33	75.96	4.44	10.83	95.07	5.15
J	95.76	7.52	1.03	1.15	24.72	65.70	2.42	5.36	92.76	9.22
K	-	7.50	1.01	1.17	-	32.45	1.51	2.40	-	4.34
Subprime 1	94.81	18.51	1.41	1.42	83.68	50.73	2.00	2.68	88.62	8.85
Subprime 2	96.93	10.41	1.20	1.33	96.48	59.07	1.75	2.95	97.38	2.98
Mean	86.77	13.09	1.18	1.36	39.23	74.18	3.14	7.92	81.01	-
Weighted Mean	79.51	14.32	1.19	1.38	32.51	77.80	3.34	9.18	78.58	-
NS Mean	84.75	12.84	1.15	1.36	27.93	77.69	3.38	8.85	78.34	-
NS Weight Mean	77.08	14.03	1.17	1.37	24.22	81.15	3.53	10.04	76.72	-

Notes: “Share” column reports share of originations; “C. of V.” columns report coefficients of variation; “75/25” and “90/10” columns report 75th to 25th and 90th to 10th percentile ratios respectively; “within” columns report the ratio of within to total variation, in percentage terms. All values are averages over months. The weighted mean is weighted by the number of originations. NS stands for “no subprime”, and NS means calculate the mean omitting the subprime lenders. Missing values of within correspond to lenders who only offer one card, so means are over lenders offering multiple cards. Lenders’ identities are scrambled for confidentiality reasons and do not necessarily match the identities in other tables and figures. Shares may not add up to 100 because of rounding.

TABLE A.3. FIRST AND SECOND STEP DEMAND ESTIMATES: GENERAL MODEL

Variable	Interpretation	Parameter	SE ($\times 10^{-2}$)
η^D	Default Equation Constant	-1.92	2.51
Ω^D	Default-Income Gradient	-0.14	2.12
σ^D	Variation in Default Unobservables	0.32	2.49
$\Omega^{B,\text{cons}}$	Revolving-Income Gradient	0.44	2.31
$\Omega^{B,r}$	Income Gradient for Revolving Sensitivity	-0.86	2.06
σ^B	S.D. in Revolving Unobservable	2.67	3.46
$\text{Corr}(\varepsilon^B, \varepsilon^D)$	Correlation in Unobservables	0.32	2.49
$\Omega^{E,r}$	Income Gradient for Card-Choice Sensitivity	-0.77	0.03
$\Omega^{E,\text{cons}}$	Transacting-Income Gradient	-0.11	1.43
ϱ	Transact/Revolve Substitution Parameter	0.31	0.33

Notes: This table reports the mean of parameters and their standard errors over markets. These estimates are from the general model, in which $\Omega_{mt}^{E,r}$ is estimated. The baseline model instead sets $\Omega_{mt}^{E,r} = 0$; see Table 1 for the baseline first- and second-step estimates.

TABLE A.4. THIRD STEP DEMAND ESTIMATES

	(1)	(2)	(3)	(4)
Dependent Variable	δ^B	δ^B	δ^E	δ^E
Price Sensitivity (α)	2.79	-0.93	1.25	-1.63
Airmiles (β_{airmiles})			0.00	0.03
Affinity (β_{affinity})			-0.05	0.01
Estimation	OLS	IV	OLS	IV
First-Stage F-Statistic	-	33.56	-	38.11

Notes: This table provides estimates of the demand parameters recovered in the third stage of demand estimation. In IV specifications, I use a cost shifter as an excluded instrument for the interest rate. I include bank and month fixed effects in both regressions, along with network and distribution fixed effects in δ^E regressions.

B Additional Descriptive Analysis

B.1 ANOVA Formulas

In this subsection, I formally describe the ANOVA methodology used in Section 3 and Section B.2 below. I decompose variation in lenders' credit scores, interest rates, and credit limits

into within-card and between-card components. For each lender ℓ and month t , I split the total variation $V_{\ell t}^{TOT}$ in outcome $y_{ij\ell t}$ across cards $j \in J_{\ell t}$ and originations $i \in I_{j\ell t}$ into within-card variation $V_{\ell t}^W$ and between-card variation $V_{\ell t}^B$. The decomposition is as follows:

$$\underbrace{\frac{1}{I_{\ell t}} \sum_{j=1}^{J_{\ell t}} \sum_{i=1}^{I_{j\ell t}} (y_{ij\ell t} - \bar{y}_{\ell t})^2}_{V_{\ell t}^{TOT}} = \underbrace{\frac{1}{I_{\ell t}} \sum_{j=1}^{J_{\ell t}} \sum_{i=1}^{I_{j\ell t}} (y_{ij\ell t} - \bar{y}_{j\ell t})^2}_{V_{\ell t}^W} + \underbrace{\sum_j s_{j\ell t} (\bar{y}_{j\ell t} - \bar{y}_{\ell t})^2}_{V_{\ell t}^B},$$

where $I_{\ell t}$ is the total number of originations at lender ℓ in month t , $\bar{y}_{\ell t}$ is the grand mean of outcome y , $\bar{y}_{j\ell t}$ is the card- j -specific mean, and $s_{j\ell t} = \frac{I_{j\ell t}}{I_{\ell t}}$ is the share of originations on card j at lender ℓ in month t . Intuitively, the decomposition separates the grand variance into an average of within-card variances ($V_{\ell t}^W$) and a weighted variance of card averages ($V_{\ell t}^B$). The R-squared from a regression of $y_{ij\ell t}$ on dummies for each of the cards j in a given lender-month pair ℓt provides the ratio $V_{\ell t}^B / V_{\ell t}^{TOT}$.

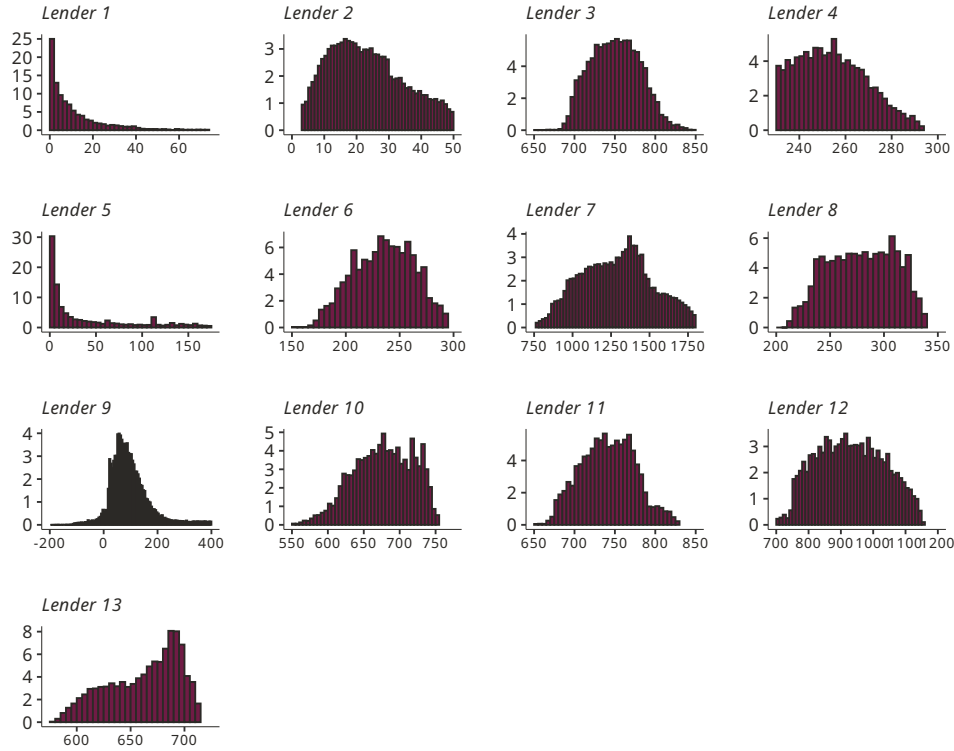
B.2 Credit Scores

I highlight two features of the credit score data: (i) credit scores vary across lenders and are based on a lender's proprietary information, and (ii) there is substantial *within-card* variation in credit scores among cardholders at origination.

In Figure B.1, I plot the distributions of origination credit scores across lenders in the year 2013, the only year in which credit scores are stable for all lenders. The figure shows that lenders' scores differ in their numerical scales and distribution shapes. These features suggest that lenders construct their own scores, but these differences could result from a mere rescaling process at each lender or customers with differing default risks selecting across different lenders.

To explore further, I regress each lender's proprietary credit scores on a fine set of demographics that lenders collect during the application process. Specifically, I include 100 equal-width

FIGURE B.1. HISTOGRAM OF PROPRIETARY CREDIT SCORES ACROSS LENDERS



Notes: I scramble lenders' identities to preserve anonymity, so labels do not necessarily match the identities in other tables and figures. The plot is constructed using 2013 data.

income bins, bins for each age, dummies for employment and homeownership status, and month fixed effects. In these regressions, the across-lender average proportion of variation in private credit scores explained is 21%, ranging from 7% to 34%.¹ These findings imply that most lenders' proprietary credit scores are based on more than the readily available customer demographics, though some lenders rely on these demographics more than others. The variation in R-squared values across lenders suggests that lenders' proprietary credit

¹Similar findings emerge when I perform the same exercise, replacing demographics with the main publicly available UK credit score, though I only have data on the public credit score for a limited set of months. In these regressions, the mean R-squared across lenders is around 22%, and it varies from 6% to 36%. These findings imply that public credit scores explain only a moderate share of the variation in each lender's proprietary credit scores: lenders either incorporate private data sources into their algorithms or use algorithms different from those of public credit score providers.

scoring algorithms differ.

Finally, I investigate whether lenders sort customers of similar risk onto different cards. For each lender-month, I run a one-way ANOVA (as detailed in Section B.1) to decompose variation in proprietary credit scores into within-card and between-card components. Table A.2 column (1) contains the values over all lenders. When averaging over lenders and months, the within-card variation accounts for 87% of the total variation, indicating substantial variation in customers' credit scores *within* each card product. This fact implies that lenders by and large *do not* segment customers of varying risk onto separate cards.

B.3 Ruling Out Alternative Forms of Risk-Based Pricing

Lenders could employ risk-based pricing by adjusting interest rates after origination, *repricing* customers based on their evolving risk and behavior. However, repricing is limited in the UK credit card market. Outside promotional deals, I calculate that lenders reprice 3% of customers within nine months of origination and 5% within one year of origination.

Furthermore, lenders could employ risk-based pricing by offering multiple distinct cards and sorting customers of differing risk onto different cards. Section B.2 shows that customers are not sorted onto cards by credit score, and Section 3.1 shows that there is limited between-card variation in interest rates at a lender. In addition, lenders offer a small set of distinct card products each month. For example, the average share of originations on the top two cards at each lender is 86% at the mean and 91% at the median.

Finally, interest rates across lenders do not move together over time, and significant price decreases by one lender are not followed by similar decreases by other lenders. These findings are consistent with the absence of collusion among lenders, but are only indicative; formally ruling out collusion in credit markets remains a matter for further research.

B.4 Default and Asymmetric Information

I follow the empirical literature on credit and insurance markets and examine the relationship between credit demand and default. My focus is the intensive margin of demand, examining the relationship between the amount of credit borrowed and failure to repay.

I estimate the correlation between unobservables that determine credit card borrowing and serious delinquencies by following the approach first suggested in [Chiappori and Salanié \(2000\)](#) and subsequently used in credit markets by [Crawford et al. \(2018\)](#) and [Benetton and Buchak \(2024\)](#). A positive correlation between unobservables driving credit demand and default can arise from “adverse selection” (those with ex-ante unobservables driving high levels of borrowing also having unobservables that increase the likelihood of failure to repay) or from positively correlated ex-post shocks to borrowing and ability to repay. While I do not attempt to separately identify these forces, any positive correlation reflects the extent of combined information asymmetries in this market.

The two estimating equations are $y_{it}^{v*} = x_{it}\beta^v + \epsilon_{it}^v$, for $v \in \{b, d\}$. The variable y_{it}^{d*} represents the latent net utility from failing to repay the credit card. Consistent with standard practice by UK credit card lenders, the observable $y_{it}^d = 1(y_{it}^{d*} > 0)$ is equal to one if consumer i who originated in month t went 90 days without making a credit card repayment in the first 18 months after t . The variable y_{it}^b is a dummy variable equal to one if the individual revolves their full credit limit. The variable x_{it} represents a flexible set of individual characteristics available to lenders. (logged income, origination month, card product dummies, and distribution channel)

I assume that ϵ_{it}^b and ϵ_{it}^d are standard normal variables to yield the Chiappori and Salanié “Pair of Probits.” I calculate the correlation between the generalized residuals $\hat{\epsilon}_{it}^b$ and $\hat{\epsilon}_{it}^d$ along with the [Chiappori and Salanié \(2000\)](#) W statistic, which is asymptotically $\chi^2(1)$ under the null of $\text{corr}(\epsilon_{it}^b, \epsilon_{it}^d) = 0$. I use an unweighted version of the [Chiappori and Salanié](#)

(2000) statistic, given by

$$W = \frac{(\sum \hat{\epsilon}_{it}^b \hat{\epsilon}_{it}^d)^2}{\sum (\hat{\epsilon}_{it}^b \hat{\epsilon}_{it}^d)^2}.$$

The correlation is 0.12 (95% confidence interval [0.119,0.123]), implying a modest but robust positive relationship between unobservables driving default and revolving. The W statistic is 9312. These findings are consistent with asymmetric information in credit card markets and motivate including correlated unobservables for borrowing and default in the model.

B.5 Credit Limit Regression Discontinuity Design

I isolate quasi-random variation in the origination credit limit at a single lender whose credit limit schedule steps up at a threshold in the application risk score. In this regression discontinuity design (RDD), the running variable is the application risk score, the cutoff is the lender’s risk score threshold, and the cohort comprises the lender’s originations in September 2013. The RDD is fuzzy: crossing the threshold raises the credit limit discontinuously but not deterministically, because the assignment mechanism includes other discretionary elements. I study three post-approval outcomes for the calendar year after origination: closing the focal card (*cancel*), opening another card (*additional*), and doing both (*switch*).

I estimate local-linear regressions with a triangular kernel and MSE-optimal bandwidths, reporting robust bias-corrected confidence intervals (Calonico et al., 2014), and I check honest confidence intervals based on a Hölder smoothness bound (Armstrong and Kolesár, 2018; Kolesár and Rothe, 2018). I assess validity using a manipulation test (Cattaneo et al., 2018; 2020), covariate continuity, bandwidth-sensitivity curves, and finite-sample randomization inference (Cattaneo et al., 2015).

I run the local polynomial density test of Cattaneo et al. (2018) to assess the design’s validity. I find no evidence of manipulation of the running variable at the cutoff ($p = 0.35$). Predetermined covariates are also continuous through the threshold, with age ($p = 0.46$) and

monthly income ($p = 0.26$) showing no significant discontinuity. Crucially, the origination interest rate shows no jump at the cutoff: the estimated difference is about 0.1 percentage points on a base of roughly 16%. It is not significant under the honest test ($p \approx 0.21$) nor the robust bias-corrected test ($p = 0.06$). The threshold therefore shifts the credit limit while leaving the price of borrowing essentially unchanged.

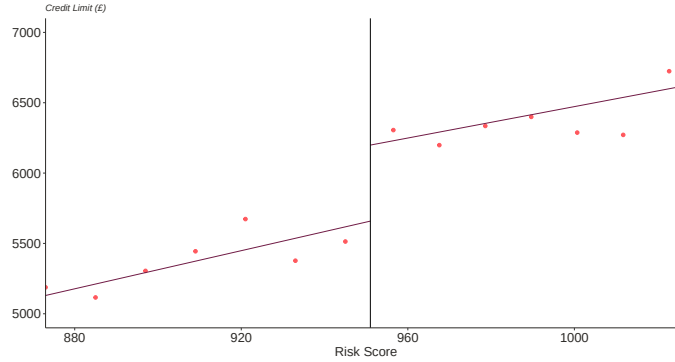
Regarding the first stage, crossing the threshold raises the origination credit limit by £754 (95% confidence interval [$\pounds 357$, $\pounds 1,281$]). Panel A of Figure B.2 shows the first-stage plot. Regarding the outcome variables, none of the three post-approval margins yield a statistically significant change at the credit limit threshold. The effect on the probability of switching is precisely zero, estimated at -0.1 percentage points (95% confidence interval $[-2.4, +1.8]$) on a base of 3.7% ($p = 0.78$). The new card margin estimate is -1.7 percentage points (95% confidence interval $[-6.2, 2.0]$) on a base of 19.6% ($p = 0.32$). The cancellation probability, on a base of 10.7%, moves by a statistically insignificant -1.5 percentage points (95% confidence interval $[-5.1, +1.9]$), with $p = 0.37$. Scaling the reduced form by the first stage, a £1,000 increase in the limit changes the switch, additional-card, and cancellation probabilities by -0.2 , -2.3 , and -2.1 percentage points, respectively, none statistically significant. The null results are stable across bandwidths from half to twice the MSE-optimal value (see panel B of Figure B.2). Finally, finite-sample randomization inference in a narrow window around the threshold yields the same conclusions.

C Details on the Model

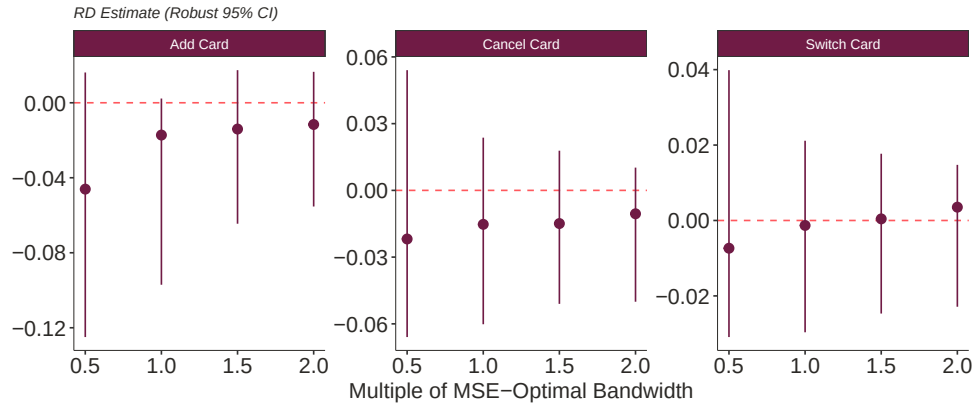
C.1 Grounds for Borrowing Depending on Advertised Rate

Two economic grounds support modeling borrowing as a function of the advertised rate. First, because individualized rates are set after the contract is signed, it is reasonable to assume that individuals choose their card and *initial* borrowing based on the advertised rate, then adjust their long-term borrowing after they learn their individualized rate. Further,

FIGURE B.2. CREDIT LIMIT RDD: FIRST-STAGE AND BANDWIDTH SENSITIVITY



PANEL A: Origination credit limits at the score threshold



PANEL B: Estimates across alternative bandwidths

individuals might commit to some spending on the card before their individualized rate is determined.

Second, this modeling choice fits the framework of [Gabaix and Laibson \(2006\)](#), with individualized rates as the shrouded object that is not salient for customers, particularly when they differ from the advertised rate. If individuals choose their card by comparing advertised interest rates, then the advertised rate will be the salient variable. And if individuals commit to their initial revolving before getting the card, they might choose their revolving based on the advertised rate.

There is an econometric justification for this choice too. Since the advertised rate is the

individualized rate less the deviation ($r_{jmt} = r_{ijmt} - z_{ijmt}$), replacing individualized rates with advertised rates in estimation is equivalent to additive measurement error contamination. Because approximately 90% of customers obtain the advertised rate, only 10% of observations are measured with error, suggesting that the quantitative impact of using the advertised rate is likely to be small.

C.2 Null Effects of Interest Rates and Credit Limits on Default

I follow the empirical literature on credit markets and omit interest rates from the default utility. [Nelson \(2025\)](#) and [Castellanos et al. \(2018\)](#) provide empirical evidence that price has an insignificant effect on default in credit markets. [Collier et al. \(2024\)](#) exploits a discontinuity in the interest rate of disaster loans and finds no difference in delinquency rates on either side of the discontinuity. Much of the research on default suggests that short-run liquidity drives default rather than a loan’s long-run value ([Ganong and Noel, 2020](#); [Indarte, 2023](#)). As well, making default invariant to price follows other structural models of selection markets without direct interest rate effects on default, for example, [Cohen and Einav \(2007\)](#).

Further, as in [Nelson \(2025\)](#), default is not a direct function of the credit limit. Empirical work supporting this choice includes (i) [Gross and Souleles \(2002\)](#), which shows that increases in credit limits do not explain increases in default, and (ii) [Agarwal et al. \(2017\)](#), which provides evidence that credit limits do not affect default rates among prime customers, which are the subset of customers I use to estimate my model.

I provide evidence from my data that corroborates the previously cited work’s conclusions on the relationship between credit limits and default. I regress a dummy for cardholder default 18 months after origination on (logged) origination credit limit, income, and market dummies, using the subset of revolvers. In all models (linear probability model, probit, and logit), the coefficient on credit limit is *negative* and strongly statistically and economically significant. Of course, this correlation alone does not preclude a positive causal effect of

credit limit on default. Instead, it is consistent with the selection effect coming from lenders endogenously choosing lower credit limits for risky customers (the supply-side effect) empirically dominating any positive demand-side causal effect. This selection effect is central to the supply-side model I estimate.

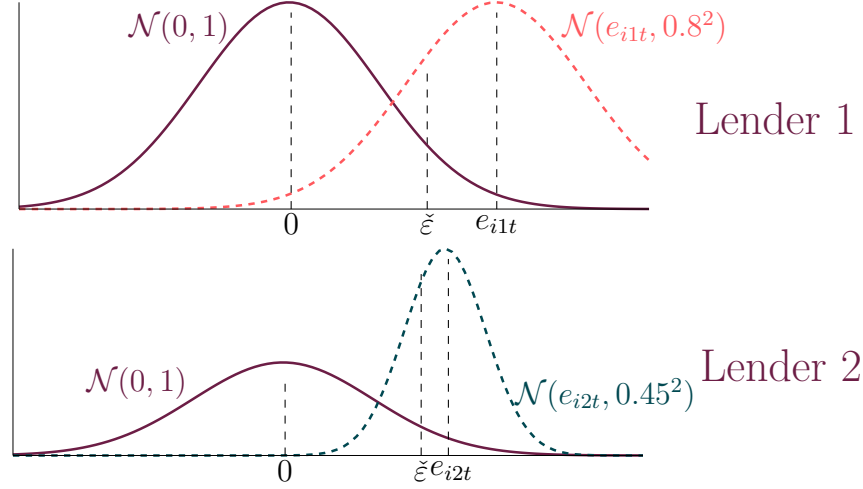
C.3 Modeling of Screening Technologies

The screening technology takes in data available to the lender about a customer and provides a tailored prediction of possible values of the customer’s common risk component ε_i . Without a screening technology, for each customer, the lender would take the expectation over a standard normal, which is the population distribution of ε_i . The screening technology intends to provide a distribution with a mean close to each individual’s realization of ε_i and a variance less than one, which is that of the population distribution.

Two features characterize the lender-specific, tailored distributions that the screening technology delivers. The first is the set of signals, or central points, around which the tailored distributions can be based. I denote these as e_{ilt} , which can take a finite number of lender-specific values $\{e_{\ell t 1}, \dots, e_{\ell t L_{\ell t}}\}$. The second feature is the precision of the distribution it generates. The distribution generated by the screening technology accounts for the possibility that the signal may not correctly predict a customer’s risk; that is, it allows for error. For an individual who generated the signal e_{ilt} , the distribution provided by the screening technology is normal with mean e_{ilt} and variance $\sigma_{\ell t}^2 \leq 1$, and I call $\sigma_{\ell t}$ the dispersion parameter (so that $1/\sigma_{\ell t}^2$ measures precision). Given the value of e_{ilt} , the lender treats the borrower’s risk as ε_i as $\hat{\varepsilon}_i = e_{ilt} + w_{ilt}$, where $w_{ilt} \sim \mathcal{N}(0, \sigma_{\ell t}^2)$. When setting profits, the lender takes expectations using the distribution $\mathcal{N}(e_{ilt}, \sigma_{\ell t}^2)$, which the screening technology provides.

Figure C.1 depicts distributions of ε_i and $\hat{\varepsilon}_i$ for two fictitious lenders. The risk distribution provided by lender 1’s screening technology for customer i is $\mathcal{N}(e_{i1t}, 0.8)$. The mean of the conditional distribution e_{i1t} is far from the customer i ’s actual realization of $\varepsilon_i = \check{\varepsilon}$. Lender

FIGURE C.1. DISTRIBUTION OF ε (SOLID) AND $\hat{\varepsilon}_i$ (DASHED) ACROSS TWO LENDERS FOR A CUSTOMER WITH UNKNOWN VALUE $\varepsilon_i = \tilde{\varepsilon}$



Notes: The bottom lender's screening technology, which delivers the signal e_{i2t} , outperforms the top lender's signal of e_{i1t} for this individual. The standard normal $\mathcal{N}(0, 1)$ is the population distribution of ε_i .

2 has a better screening technology. Its screening technology gives the signal e_{i2t} , which is closer to $\tilde{\varepsilon}$. Furthermore, since σ_2 is smaller than σ_1 , the signal errors at lender 2 are less dispersed around the signal than at lender 1. When setting credit limits for customer i , lender 2 will put more weight (relative to lender 1) on potential values close to $\tilde{\varepsilon}$ and less on incorrect values, such as those near zero.

How does this model relate to alternatives in the literature? Unlike work that endogenizes scorecard coarseness as a lender's choice (Livshits et al., 2016; Agarwal et al., 2017) or that makes heterogeneous screening the central object of study (Cherry, 2024), this model treats a lender's technology as an exogenous building block rather than a focus of the analysis. Other recent work on screening technologies includes Blattner et al. (2022), He et al. (2024), Farboodi et al. (2025), and Blattner and Nelson (2026).

C.4 Focus on Interest Revenue

I focus on interest revenue in lenders' revenue because it comprises the majority of revenue for US lenders, specifically around 70% (Evans and Schmalensee, 2005). The remaining 30% includes alternative revenue sources that likely make up smaller proportions of total revenue in the UK than in the US. I describe the three largest alternative revenue sources below.

The first is **interchange revenue**, which accounts for 15% of US lenders' revenues on average. Interchange revenue is the money lenders receive from merchants and their banks when individuals use their cards for purchases. Between 2010 and 2015, interchange fees were significantly lower in the UK than in the US, likely resulting in a smaller share of UK lenders' revenue from interchange fees.²

The second part of the remaining 30% of non-interest revenue comes from **cash advances**. Cash-advance fees are the charges consumers pay for using a credit card to withdraw cash or conduct other non-standard card activities such as gambling. Cash-advance revenues became a negligible part of UK lenders' revenue in April 2011, when new credit card regulation forced lenders to start applying customers' repayments to high-interest cash advance balances first.

The final source of revenue is **fee revenue**. Around 88% of cards in the UK have no annual fee, so I focus on other fees. All other fees were reduced because of a UK policy investigation between 2003 and 2006. In 2003, the Office of Fair Trading (OFT) began an inquiry into the "default charges" levied by credit card companies when, for example, a cardholder exceeded their credit limit or was late making the minimum monthly payment. In 2006, the OFT stated that many of the charges were "unlawful," saying it would act upon receiving notice of any fee over £12 (Office of Fair Trading, 2006). In 2010–2015, all fees apart from annual fees (including late, dormancy, over-limit, and foreign transaction) were £12, around 50%

²The European Parliament and the Council of the European Union adopted the Interchange Fee Regulation (IFR), which set the default interchange fee cap at 0.3% of the transaction for credit cards. See Wang (2023) for more discussion of merchant fee caps.

lower than in 2003 (House of Commons Treasury Committee, 2003). Fees in the US are levied more frequently and are usually larger than £12, further suggesting that fees accounted for a smaller proportion of UK lenders’ revenues.

C.5 Potential Explanations for the Lack of Risk-Based Pricing

Interviews with industry experts identified two candidate explanations for the lack of risk-based pricing in the UK, both of which can be viewed as costs of individualizing interest rates. First, as described in Section 2, EU regulations require that at least 51% of customers who originate a card must receive the advertised APR or lower. This constraint directly impedes lenders in fully individualizing prices. Because the regulatory constraint limits the number of customers over which a lender could recover the high fixed costs of building automated pricing infrastructure (e.g., software, data modeling), lenders may have opted against making those large setup investments. Consequently, implementing individual rate deviations within their existing, less flexible systems imposes high variable administrative and operational frictions for each deviation. The cost parameter I estimate, κ_{ijmt} , captures these high shadow costs of individualizing rates under the current infrastructure, rather than the unobserved fixed setup costs themselves. Given these restrictions and resulting frictions, lenders might have focused their investments on individualized credit limits instead.

Second, lenders may incur high reputational costs if they advertise a particular APR but then provide customers with a different, individualized APR, especially because the individualized rate is set after the individual signs the contract. In fact, members of the UK Government expressed their disapproval of such practices (House of Commons Treasury Committee, 2003). This issue is a focal point for lenders, who recognize that negative attention stemming from unpopular business practices generates reputational risk. A substantial body of literature underscores the importance of reputational risk in the banking sector (see, e.g., Fiordelisi, Soana, and Schwizer, 2013 and Scandizzo, 2011). My dataset covers the years immediately following the global financial crisis—an event that significantly impaired the public’s atti-

tude towards the banking industry (Bennett and Rita, 2012). Therefore, in the short term, avoiding further reputational damage was likely a primary objective for credit card lenders.

C.6 Credit Limit First Order Condition Derivation

In this subsection, I derive Equation (7) from the first order condition of the lender's profit maximization problem. The first step is to replace ε_i with $e_{ilt} + w_{ilt}$. The second step is to note that for every $\bar{b}$, there exists a threshold signal error $\omega_{ilt}(\bar{b})$ such that if the signal error w_{ilt} is larger (respectively smaller) than ω_{ilt} , the individual's desired borrowing will be larger (respectively smaller) than $\bar{b}$.³ The value of ω_{ilt} sets $\log(b_{ijmt}^*)$ equal to $\log(\bar{b}_{ijmt})$ and is therefore given by

$$\omega_{ilt}(\bar{b}_{ijmt}, e_{ilt}) = \frac{\log(\bar{b}_{ijmt}) - \delta_{jmt}^B - u_{ijmt}^B}{\sigma_{mt}^B} - e_{ilt}.$$

From this, I split Π into

$$\frac{1}{\sigma_{\ell t}} \int_{-\infty}^{\omega_{ilt}} b_{ijmt}^* \pi_{ijmt}(e_{ilt}, w_{ilt}) \phi\left(\frac{w_{ilt}}{\sigma_{\ell t}}\right) dw_{ilt} + \frac{\bar{b}_{ijmt}}{\sigma_{\ell t}} \int_{\omega_{ilt}}^{\infty} \pi_{ijmt}(e_{ilt}, w_{ilt}) \phi\left(\frac{w_{ilt}}{\sigma_{\ell t}}\right) dw_{ilt}. \quad (\text{C.1})$$

By the Leibniz integral rule, the first derivative with respect to $\bar{b}_{ijmt}$ is equal to

$$\frac{1}{\sigma_{\ell t}} \int_{\omega_{ilt}}^{\infty} \pi_{ijmt}(e_{ilt}, w_{ilt}) \phi\left(\frac{w_{ilt}}{\sigma_{\ell t}}\right) dw_{ilt} \quad (\text{C.2})$$

and the second derivative

$$-\frac{1}{\sigma_{\ell t}} \frac{d\omega_{ilt}}{d\bar{b}_{ijmt}} \pi(e_{ilt}, \omega_{ilt}) \phi\left(\frac{\omega_{ilt}}{\sigma_{\ell t}}\right) = -\frac{1}{\sigma_{\ell t} \sigma_{mt}^B \bar{b}_{ijmt}} \pi(e_{ilt}, \omega_{ilt}) \phi\left(\frac{\omega_{ilt}}{\sigma_{\ell t}}\right),$$

³This version assumes that σ_{mt}^B is positive, a condition I impose in estimation without loss of generality. The sign of σ_{mt}^B is not identified, so I normalize it as positive. If I normalize σ_{mt}^B as negative, the first order condition bounds would swap to $(-\infty, \omega_{ilt}]$, but the equation is otherwise unchanged.

which is negative provided that $\pi(e_{ilt}, \omega_{ilt}) > 0$. In this region, the objective is concave, and the first order condition is necessary and sufficient for a maximum. Finally, given that the second order condition requires $\pi(e_{ilt}, \omega_{ilt}) > 0$ and the integral runs from ω_{ilt} to ∞ , for the first derivative to equal zero, π should decrease in w_{ilt} , which requires $\sigma_{mt}^D > 0$. I estimate the parameter $\sigma_{mt}^D > 0$ as part of the demand side estimation, which is completely independent of the supply side estimation. The estimates confirm that the required sign restriction holds in my context.

C.7 Formulas for Elasticities of Demand

I derive formulas for the demand elasticities for both the borrowing level b_{ijmt} and extensive product choice. I start with the borrowing level. The elasticity for individual i is

$$\frac{\partial \log(b_{ijmt})}{\partial \log(r_{jmt})} = r_{jmt} \frac{\partial \log(b_{ijmt})}{\partial r_{jmt}}.$$

In the econometric model, the right-hand side derivative is the marginal effect from a Tobit model with top censoring at $\log(\bar{b}_{ijmt})$. As shown in [Greene \(2017\)](#), the marginal effect on expected borrowing in this econometric model is

$$\alpha_{imt}^B \Phi \left(\frac{\bar{Q}_{ijmt}^B}{\sigma_{mt}^B} \right),$$

where $\bar{Q}_{ijmt}^B$ is defined in Equation (D.4). From this, the elasticity of expected borrowing is

$$\frac{\partial \mathbb{E}[\log(b_{ijmt})]}{\partial \log(r_{jmt})} = r_{jmt} \alpha_{imt}^B \Phi \left(\frac{\bar{Q}_{ijmt}^B}{\sigma_{mt}^B} \right). \quad (\text{C.3})$$

The elasticity for the extensive product choice is more involved. By definition, the probability that an individual chooses card j as a borrower is

$$s_{ijmt}^E = (1 - s_{i0mt}^E) s_{ijmt|j \in J_{imt}}^E,$$

where $s_{ijmt|j \in J_{imt}}^E$ is the probability of individual i choosing card j , conditional on revolving (see Equation (D.2)), and s_{i0mt}^E is the probability that individual i chooses to transact. From this,

$$\frac{\partial s_{ijmt}^E}{\partial r_{jmt}} = (1 - s_{i0mt}^E) \frac{\partial s_{ijmt|j \in J_{imt}}^E}{\partial r_{jmt}} - s_{ijmt|j \in J_{imt}}^E \frac{\partial s_{i0mt}^E}{\partial r_{jmt}}.$$

The standard logit derivative for the inside options is

$$\frac{\partial s_{ijmt|j \in J_{imt}}^E}{\partial r_{jmt}} = s_{ijmt|j \in J_{imt}}^E (1 - s_{ijmt|j \in J_{imt}}^E) \frac{\alpha_{imt}^E}{\varrho_{mt}}$$

and the derivative of the outside option probability is

$$\frac{\partial s_{i0mt}^E}{\partial r_{jmt}} = -\alpha_{imt}^E s_{ijmt|j \in J_{imt}}^E s_{i0mt}^E (1 - s_{i0mt}^E) = -\alpha_{imt}^E s_{i0mt}^E s_{ijmt}^E.$$

Putting these together yields

$$\frac{\partial s_{ijmt}^E}{\partial r_{jmt}} = \alpha_{imt}^E s_{ijmt}^E \left[\frac{1 - s_{ijmt|j \in J_{imt}}^E}{\varrho_{mt}} + s_{ijmt|j \in J_{imt}}^E s_{i0mt}^E \right]. \quad (\text{C.4})$$

Multiplying (C.4) by $\frac{r_{jmt}}{s_{ijmt}^E}$ provides the product choice price elasticity of demand for individual i , given by

$$\frac{\partial \log(s_{ijmt}^E)}{\partial \log(r_{jmt})} = r_{jmt} \alpha_{imt}^E \left[\frac{1 - s_{ijmt|j \in J_{imt}}^E}{\varrho_{mt}} + s_{ijmt|j \in J_{imt}}^E s_{i0mt}^E \right].$$

D Details on Estimation

D.1 First Step

The term in the log-likelihood for card choice parameters is

$$\log \mathcal{L}_{mt,E} = \sum_{i \in I_{mt}} \sum_{j \in J_{imt}} 1_{ijmt}^E \log(s_{ijmt|j \in J_{imt}}^E), \quad (\text{D.1})$$

where $1_{ijmt}^E = 1(j_{imt}^* = j)$ is a dummy equal to one if individual i chooses card j in their choice set J_{imt} and $s_{ijmt|j \in J_{imt}}^E$ reflects the likelihood that individual i chooses card j in channel m and origination month t , *conditional* on individual i choosing to revolve a credit card balance. It is given by

$$s_{ijmt|j \in J_{imt}}^E = \frac{\exp(\bar{U}_{ijmt}^E)}{\sum_{k \in J_{imt}} \exp(\bar{U}_{ikmt}^E)}, \quad (\text{D.2})$$

where scaled indirect utility $\bar{U}_{ijmt}^E$ is $\bar{U}_{ijmt}^E = \frac{\bar{V}_{ijmt}^E}{\varrho_{mt}}$.

Next, I provide the joint log-likelihood expression for revolving and default choices, $\log \mathcal{L}_{mt,BD}$. Revolvers either use their entire credit line (full utilization) or not (interior utilization) and do or do not default. This creates four possible outcomes for revolver i :

1. $i \in I_1$: Interior utilization and default
2. $i \in I_2$: Interior utilization and no default
3. $i \in I_3$: Full utilization and default
4. $i \in I_4$: Full utilization and no default

Let $s_{ijmt}^{(g)}$ denote the likelihood of individual i being in group I_g . Then

$$\log \mathcal{L}_{mt,BD} = \sum_{i \in I_{mt}} \sum_{j \in J_{imt}} \sum_{g=1}^4 1_{ijmt}^{(g)} \log(s_{ijmt}^{(g)}), \quad (\text{D.3})$$

where $1_{ijmt}^{(g)}$ is a dummy equal to one if individual i chooses card j and is in group I_g . The expressions for $s_{ijmt}^{(g)}$, $g = 1, \dots, 4$ are given as follows. The first term $s_{ijmt}^{(1)}$ for an individual who borrows $b < \bar{b}_{ijmt}$ and defaults is

$$\begin{aligned} s_{ijmt}^{(1)} &= \mathbb{P}(\text{Default} | \log(b_{ijmt}^*) = \log(b)) \cdot f_{\log(b_{ijmt}^*)}(\log(b)) \\ &= \frac{1}{\sigma_{mt}^B} \mathbb{P}(\varepsilon_{imt}^D > -\mathcal{Q}_{imt}^D | \varepsilon_{imt}^B = \mathcal{Q}_{ijmt}^B(b)) \phi\left(\frac{\mathcal{Q}_{ijmt}^B(b)}{\sigma_{mt}^B}\right) \\ &= \frac{1}{\sigma_{mt}^B} \Phi_{ijmt}^{BD,1} \phi\left(\frac{\mathcal{Q}_{ijmt}^B(b)}{\sigma_{mt}^B}\right), \end{aligned}$$

where

$$\begin{aligned}\Phi_{ijmt}^{BD,1} &= \Phi \left(\mathcal{Q}_{imt}^D + \frac{\sigma_{mt}^D}{\sigma_{mt}^B} \mathcal{Q}_{ijmt}^B(b) \right) \\ \mathcal{Q}_{ijmt}^B(b) &= \log(b) - \delta_{jmt}^B - u_{ijmt}^B, \\ \mathcal{Q}_{imt}^D &= \eta_{mt}^D + \Omega_{mt}^D \tilde{y}_{imt},\end{aligned}$$

By a similar derivation,

$$s_{ijmt}^{(2)} = \frac{1}{\sigma_{mt}^B} \left[1 - \Phi_{ijmt}^{BD,1} \right] \phi \left(\frac{\mathcal{Q}_{ijmt}^B(b)}{\sigma_{mt}^B} \right).$$

The third and fourth terms are more complicated because of the full utilization of the credit limit. The third term $s_{ijmt}^{(3)}$ is

$$\begin{aligned}s_{ijmt}^{(3)} &= \mathbb{P}(\log(b_{ijmt}^*) > \log(\bar{b}_{ijmt})) \mathbb{P}(V_{imt}^D > 0 | \log(b_{ijmt}^*) > \log(\bar{b}_{ijmt})) \\ &= \mathbb{P}(\varepsilon_{imt}^B > \bar{\mathcal{Q}}_{ijmt}^B) \mathbb{P}(\varepsilon_{imt}^D > -\mathcal{Q}_{imt}^D | \varepsilon_{imt}^B > \bar{\mathcal{Q}}_{ijmt}^B) \\ &= \mathbb{P}(\varepsilon_{imt}^B > \bar{\mathcal{Q}}_{ijmt}^B) \int_{\bar{\mathcal{Q}}_{ijmt}^B}^{\infty} \mathbb{P}(\varepsilon_{imt}^D > -\mathcal{Q}_{imt}^D | \varepsilon_{imt}^B = a) f_{\varepsilon_{imt}^B | \varepsilon_{imt}^B > \bar{\mathcal{Q}}_{ijmt}^B}(a | \varepsilon_{imt}^B > \bar{\mathcal{Q}}_{ijmt}^B) da \\ &= \frac{1}{\sigma_{mt}^B} \int_{\bar{\mathcal{Q}}_{ijmt}^B}^{\infty} \Phi \left(\mathcal{Q}_{imt}^D + \frac{\sigma_{mt}^D}{\sigma_{mt}^B} a \right) \phi \left(\frac{a}{\sigma_{mt}^B} \right) da = \int_{\bar{\mathcal{Q}}_{ijmt}^B / \sigma_{mt}^B}^{\infty} \Phi(\mathcal{Q}_{imt}^D + \sigma_{mt}^D \tilde{a}) \phi(\tilde{a}) d\tilde{a},\end{aligned}$$

where

$$\bar{\mathcal{Q}}_{ijmt}^B = \mathcal{Q}_{ijmt}^B(\bar{b}_{ijmt}). \quad (\text{D.4})$$

Similarly,

$$s_{ijmt}^{(4)} = \int_{\bar{\mathcal{Q}}_{ijmt}^B / \sigma_{mt}^B}^{\infty} \left[1 - \Phi(\mathcal{Q}_{imt}^D + \sigma_{mt}^D \tilde{a}) \right] \phi(\tilde{a}) d\tilde{a}.$$

The first step yields estimates of the following parameters

$$\frac{\delta_{jmt}^E}{\varrho_{mt}}, \frac{\Omega_{mt}^{E,r}}{\varrho_{mt}}, \delta_{jmt}^B, \Omega_{mt}^{B,r}, \Omega_{mt}^{B,\text{cons}}, \Omega_{mt}^D, \eta_{mt}^D, \sigma_{mt}^B, \sigma_{mt}^D.$$

D.2 Second Step

An individual transacts if the utility from transacting V_{i0mt}^E exceeds the maximal utility from revolving a balance. Based on the type-1 extreme value assumption, the likelihood that this occurs for individual i is

$$s_{i0mt}^E = \frac{1}{1 + \exp(\varrho_{mt} F_{imt} - \bar{V}_{i0mt})},$$

where F_{imt} is the inclusive value given by

$$F_{imt} = \log \sum_{k \in J_{imt}} \exp(\bar{U}_{ikmt}^E)$$

and $\bar{V}_{i0mt} = \delta_{0mt} + \Omega_{mt}^{E, \text{cons}} \tilde{y}_{imt}$. Let ζ_{imt} be a dummy equal to one if the individual chooses to transact. Then the log likelihood for transacting is

$$\log \mathcal{L}_{mt}^{tr} = \sum_{i \in I_{mt}} \zeta_{imt} \log(s_{i0mt}^E) + (1 - \zeta_{imt}) \log(1 - s_{i0mt}^E).$$

Maximizing $\log \mathcal{L}_{mt}^{tr}$ market-by-market provides estimates of δ_{0mt} , ϱ_{mt} and $\Omega_{mt}^{E, \text{cons}}$, from which I recover estimates of $\Omega_{mt}^{E, r}$ and δ_{jmt}^E from the first-step estimates. Standard errors are available for the first- and second-step likelihood estimates, where they follow directly from the likelihood. For parameters and counterfactuals, I report point estimates.

D.3 Institutional Background and Timeline of PPI Case

As an instrument for interest rates, I exploit a 2011 cost shock to lenders stemming from the mis-selling of payment protection insurance (PPI). PPI is insurance designed to cover loan repayments if an individual cannot make them due to adverse events such as unemployment, illness, or disability. In the late 20th century, UK lenders began bundling PPI with credit products such as credit cards. In the mid-2000s, it was alleged that lenders were mis-selling

PPI to borrowers. For example, lenders sold PPI to self-employed individuals who could not use it because of their employment status. In 2006, the Financial Services Authority (FSA) began fining financial institutions for mis-selling PPI.

A significant development came in January 2011, when the British Bankers' Association (BBA) took the FSA to court over its decision to impose standards on the correct selling of PPI *retrospectively*.⁴ The BBA lost the case, and in mid-2011, banks told the BBA they were withdrawing their support for an appeal of the decision. The ruling forced banks to reopen thousands of PPI mis-selling claims. Around 64 million policies were mis-sold between the 1970s and 2000s, with over £38bn repaid to individuals who complained about a PPI sale. The court case loss in mid-2011 and the reopening of PPI claims led to cost increases, which were spread unevenly among banks according to how frequently they mis-sold PPI. Shortly after, some credit card lenders increased advertised interest rates for some cards in their portfolios.⁵

D.4 Estimation of Screening Technologies

For each unique observed credit limit $\bar{b}_{ijmt}$ on card j at lender ℓ in month t , the corresponding signal e_{ilt} satisfies the first order condition in Equation (7), which is equivalent to

$$\int_{\omega_{ilt}(\bar{b}_{ijmt}, e_{ilt})}^{\infty} \pi_{ijmt}(e_{ilt}, w_{ilt}) \phi\left(\frac{w_{ilt}}{\sigma_{lt}}\right) dw_{ilt} = 0, \quad (\text{D.5})$$

where I define π_{ijmt} in Equation (4). Towards an estimation strategy, note that under the distributional assumptions on private characteristics, the probability of default, as featured

⁴See *R (on the application of the British Bankers' Association) v Financial Services Authority and another [2011] EWHC 999*.

⁵Previous work argued that in the US, credit card rates are sticky relative to the cost of funds (Ausubel, 1991). However, this doesn't seem to be the case in the time series or cross-section in the UK, at least in the period for which I have data. Regarding the former, monthly average funding rates are close to 10% of interest rates. And they are not merely trending together: when I regress logged interest rates on logged funding costs along with distribution channel and *month* fixed effects, the elasticity of interest rates with respect to funding costs is 0.16 and significant at a lower than 1% significance level.

in π_{ijmt} is given by

$$\Delta_{imt} = \Phi \left(\eta_{mt}^D + \Omega_{mt}^D \tilde{y}_{imt} + \sigma_{mt}^D (e_{ilt} + w_{ilt}) \right).$$

From this expression, I can calculate Δ_{imt} and hence π_{ijmt} , and therefore the integrand, as a function of the model parameters and the signal error.

For each observed credit limit and income, Equation (D.5) yields an expression in which the only unknowns are the screening technology parameters e_{ilt} and $\sigma_{\ell t}$ (once demand parameters have been replaced with their estimates). The basis of the strategy is to estimate the screening technologies as the values that minimize the sum of squared deviations (over individuals) of the integral in (D.5). The integral in (D.5) has no closed form. Therefore, for each lender-month, I simulate the integral using H Halton draws v_{ilt}^h , and solve

$$\min \sum_{i \in I_{\ell t}} \left(\frac{1}{H} \sum_{h=1}^H 1(\sigma_{\ell t} v_{ilt}^h > \omega_{ilt}(\bar{b}_{ijmt}, e_{ilt})) \pi_{ijmt}(e_{ilt}, \sigma_{\ell t} v_{ilt}^h) \right)^2,$$

While I could estimate the model at the lender-month level, I prefer a parsimonious model that pools over all months, so that $\sigma_{\ell t} = \sigma_{\ell}$.

D.5 Marginal Costs of Individualizing Interest Rates

I derive the expression for $\frac{\partial \Pi_{ijmt}}{\partial z_{ijmt}}$ by differentiating expression (C.1) with respect to z_{ijmt} .

From the Leibniz integral rule it follows that

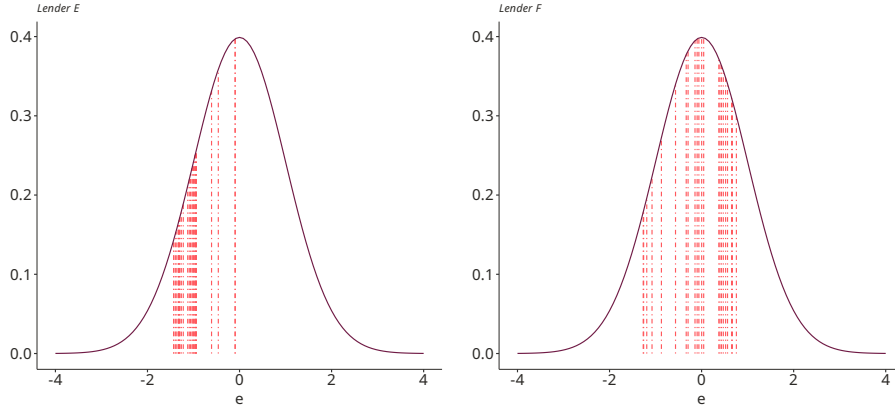
$$\frac{1}{\sigma_{\ell}} \int_{-\infty}^{\omega_{i\ell}} b_{ij}^* (1 - \Delta_i) \phi \left(\frac{w_{i\ell}}{\sigma_{\ell}} \right) dw_{i\ell} + \frac{\bar{b}_{ij}}{\sigma_{\ell}} \int_{\omega_{i\ell}}^{\infty} (1 - \Delta_i) \phi \left(\frac{w_{i\ell}}{\sigma_{\ell}} \right) dw_{i\ell}. \quad (\text{D.6})$$

As with screening technologies, these integrals can be simulated using Halton draws, allowing the derivative to be approximated. Note that if it were possible to estimate the demand side with initial borrowing depending on deviations from the advertised rate, the first term in the first integrand would become $b_{ij}^* (1 - \Delta_i) + \alpha_i^B b_{ij}^* \pi_{ij}$.

E Further Results From Supply Estimation

First, I describe the estimates of the screening technology signals, e_ℓ . Figure E.1 shows the estimated screening technologies for two contrasting lenders superimposed onto a standard normal distribution. Each vertical line represents one of the lender's signals. The right lender (lender F) has a screening technology that produces several possible signals. It is a sophisticated screening technology that provides many different signals about borrowers' types. Lender E on the left side has a screening technology that offers fewer values, concentrated on low ε realizations, implying coarser signals on risky borrowers' unobservables. These visualizations are shown for all other lenders in Figure E.2.

FIGURE E.1. SCREENING TECHNOLOGY AT TWO LENDERS

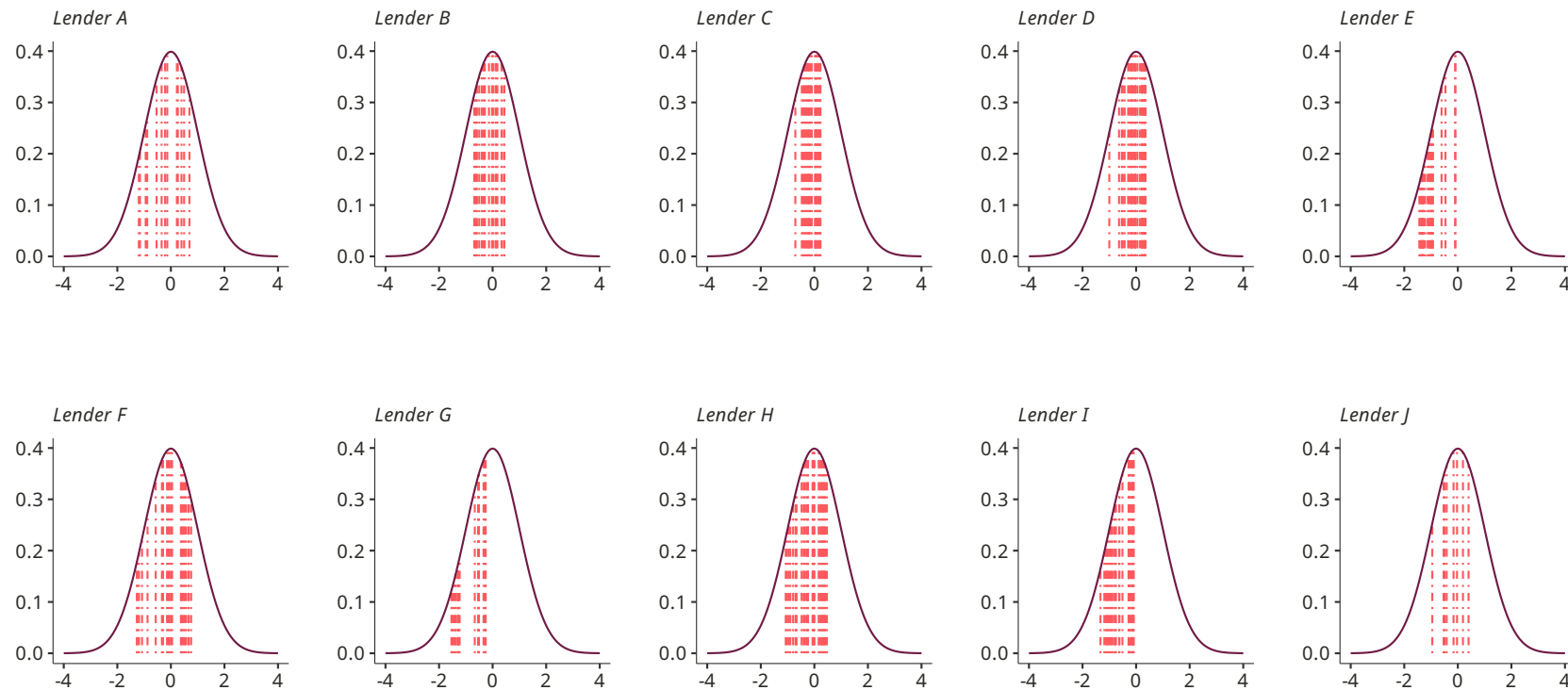


Notes: Each vertical line marks one of a lender's estimated screening signals e_ℓ , shown against the standard normal population distribution of the risk component ε that the signals partition. I scramble lenders' identities to preserve anonymity, so labels do not necessarily match the identities in other tables and figures.

Next, I describe the estimates of the costs of individualizing interest rates. In the text, I describe the distribution of these values across individuals. There are also differences across lenders in their customers' cost of individualizing interest rates. Estimates of the lender-average of κ vary from 8,926 to 24,366. The averages are largest at lenders associated with

more sophisticated, higher-education, and higher-income customer bases. These differences are statistically significant, as confirmed by regressions of κ_{ijmt} on lender and market dummies. Finally, I find that individuals with the highest income have the largest κ_{ijmt} values. These results align with what we would expect if higher-income, lower-risk individuals are either more aware of—or more aggrieved by—receiving an interest rate higher than advertised.

FIGURE E.2. SCREENING TECHNOLOGIES AT ALL LENDERS



Notes: Each vertical line marks one of a lender's estimated screening signals e_ℓ , shown against the standard normal population distribution of the risk component ε that the signals partition. I scramble lenders' identities to preserve anonymity, so labels do not necessarily match the identities in other tables and figures. Two subprime lenders and store cards are not included in the model.

F Details on Counterfactuals

In this section, I state the optimization problems for the three counterfactual scenarios I examine, with market mt fixed (and thus dropped from the notation).

F.1 Individual-Level Interest Rates and Credit Limits

For customer i , given other lenders' optimal interest rate choices $\mathbf{r}_{-i\ell}^*$, lender ℓ solves

$$\max_{\{\mathbf{r}_{i\ell}, \bar{b}_{i\ell}\}} \sum_{j \in J_{i\ell}} s_{ij}^E(\mathbf{r}_{i\ell}, \mathbf{r}_{-i\ell}^*) \mathbb{E} [\min\{b_{ij}^*, \bar{b}_{ij}\} \pi_{ij}], \quad (\text{F.1})$$

where $J_{i\ell}$ is the set of cards offered by lender ℓ with income thresholds lower than Y_i . Towards deriving the first order conditions of the optimization problem in Equation (F.1), first I define the term $\mathcal{E}_{ij}^{\text{fr}, \text{fcl}} = \mathbb{E} [\min\{b_{ij}^*, \bar{b}_{ij}\} \pi_{ij}]$. Since $\bar{b}_{ij}$ only affects the lenders' profit for card j , the first order condition with respect to $\bar{b}_{ij}$, after cancelling $s_{ij}^E(\mathbf{r}_{i\ell}, \mathbf{r}_{-i\ell}^*) > 0$, is

$$\frac{\partial \mathcal{E}_{ij}^{\text{fr}, \text{fcl}}}{\partial \bar{b}_{ij}} = 0. \quad (\text{F.2})$$

The first order condition with respect to r_{ij} is

$$\frac{\partial s_{ij}^E}{\partial r_{ij}} \mathcal{E}_{ij}^{\text{fr}, \text{fcl}} + s_{ij}^E \frac{\partial \mathcal{E}_{ij}^{\text{fr}, \text{fcl}}}{\partial r_{ij}} + \sum_{k \neq j} \frac{\partial s_{ik}^E}{\partial r_{ij}} \mathcal{E}_{ik}^{\text{fr}, \text{fcl}} = 0. \quad (\text{F.3})$$

Equation (C.4) provides an expression for $\frac{\partial s_{ij}^E}{\partial r_{ij}}$ (with r_{ij} replacing r_{jmt}). It remains to provide expressions for $\frac{\partial \mathcal{E}_{ij}^{\text{fr}, \text{fcl}}}{\partial r_{ij}}$ and $\frac{\partial s_{ik}^E}{\partial r_{ij}}$ when $k \neq j$. The former of these two terms is

$$\frac{\partial \mathcal{E}_{ij}^{\text{fr}, \text{fcl}}}{\partial r_{ij}} = \frac{1}{\sigma_\ell} \int_{-\infty}^{\omega_{i\ell}} [b_{ij}^* (1 - \Delta_i) + \alpha_i^B b_{ij}^* \pi_{ij}] \phi\left(\frac{w_{i\ell}}{\sigma_\ell}\right) dw_{i\ell} + \frac{\bar{b}_{ij}}{\sigma_\ell} \int_{\omega_{i\ell}}^{\infty} (1 - \Delta_i) \phi\left(\frac{w_{i\ell}}{\sigma_\ell}\right) dw_{i\ell}.$$

The expression for $\frac{\partial s_{ik}^E}{\partial r_{ij}}$ is given by $\frac{\partial s_{ik}^E}{\partial r_{ij}} = s_{ij}^E s_{ik|k \in J_i}^E \alpha_i^E \left[s_{i0}^E - \frac{1}{\varrho} \right]$.

The left-hand sides of equations (F.2) and (F.3) are minimized together, *for each individual separately*, to obtain the counterfactual interest rates and credit limit schedules. This means I solve thousands of optimization problems, each over $2J$ variables, which is feasible though computationally costly.

F.2 Card-level Interest Rates and Credit Limits

Given other lenders' optimal card-level interest rates $\mathbf{r}_{-\ell}^*$, lender ℓ solves

$$\max_{\{\mathbf{r}_\ell, \bar{\mathbf{b}}_\ell\}} \sum_i \sum_{j \in J_{i\ell}} s_{ij}^E(\mathbf{r}_\ell, \mathbf{r}_{-\ell}^*) \mathcal{E}_{ij}^{\text{cr}, \text{ccl}}, \quad (\text{F.4})$$

where $\mathcal{E}_{ij}^{\text{cr}, \text{ccl}} = \mathbb{E} [\min\{b_{ij}^*, \bar{b}_j\} \pi_{ij}]$ and π_{ij} has r_{ij} replaced by r_j . The first order condition with respect to $\bar{b}_j$ is

$$\sum_i s_{ij}^E \frac{\partial \mathcal{E}_{ij}^{\text{cr}, \text{ccl}}}{\partial \bar{b}_j} = 0. \quad (\text{F.5})$$

The first order condition with respect to r_j is

$$\sum_i \left[\mathcal{E}_{ij}^{\text{cr}, \text{ccl}} \frac{\partial s_{ij}^E}{\partial r_j} + s_{ij}^E \frac{\partial \mathcal{E}_{ij}^{\text{cr}, \text{ccl}}}{\partial r_j} + \sum_{k \neq j} \frac{\partial s_{ik}^E}{\partial r_j} \mathcal{E}_{ik}^{\text{cr}, \text{ccl}} \right] = 0. \quad (\text{F.6})$$

The left-hand sides of equations (F.5) and (F.6) are minimized together, *in a single step*, to solve for optimal card-level interest rates and credit limits. This case involves just one optimization problem over $2J$ variables.

F.3 Individual-Level Interest Rates and Card-Level Credit Limits

For customer i , given other lenders' individual-level interest rates $\mathbf{r}_{-i\ell}^*$, lender ℓ solves

$$\max_{\{\mathbf{r}_{i\ell}, \bar{\mathbf{b}}_\ell\}} \sum_i \sum_{j \in J_{i\ell}} s_{ij}^E(\mathbf{r}_{i\ell}, \mathbf{r}_{-i\ell}^*) \mathcal{E}_{ij}^{f, \text{ccl}}, \quad (\text{F.7})$$

where $\mathcal{E}_{ij}^{fr,ccl} = \mathbb{E} [\min\{b_{ij}^*, \bar{b}_j\} \pi_{ij}]$ with π_{ij} as a function of r_{ij} and not r_j . The first order condition with respect to r_{ij} is

$$\frac{\partial s_{ij}^E}{\partial r_{ij}} \mathcal{E}_{ij}^{fr,ccl} + s_{ij}^E \frac{\partial \mathcal{E}_{ij}^{fr,ccl}}{\partial r_{ij}} + \sum_{k \neq j} \frac{\partial s_{ik}^E}{\partial r_{ij}} \mathcal{E}_{ik}^{fr,ccl} = 0, \quad (\text{F.8})$$

and with respect to $\bar{b}_j$ is

$$\sum_i s_{ij}^E \frac{\partial \mathcal{E}_{ij}^{fr,ccl}}{\partial \bar{b}_j} = 0. \quad (\text{F.9})$$

Ideally, the left-hand sides of equations (F.8) and (F.9) would be minimized jointly over the J card-level credit limits and IJ interest rates (J interest rate offers for each of the I total individuals). Yet this is infeasible in practice – because there are first order conditions for *card-level* credit limits at optimal individual-level interest rates, the optimization cannot be performed individual-by-individual; instead, it must be performed jointly across all individuals, which results in an infeasible number of variables ($J + IJ$) to optimize over.

To address this practical issue, I instead solve an approximation of the problem, using the card-level credit limits from Section F.2, which are then plugged into Equation (F.8) to enable me to minimize the left-hand side of Equation (F.8) for each individual separately.

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